Forecasting Household Month End Electricity Bills from Smart Meter Data

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1 Problem Statement

Households usually learn what a month of electricity cost only when the bill arrives, by which point they can no longer change the usage behind it. During the month they can see what they have used so far, but not where the month's total is heading, so an expensive month is discovered too late to act on.

Smart meters record household consumption in fine detail, yet research using this data has focused on the power grid, through load analysis, forecasting and management [1], rather than on helping households anticipate their own costs.

This project investigates that gap using London households with smart meters. The problem is stated in measurable terms as follows: on any day d of a billing month, how accurately can a household's month end electricity bill be estimated from its usage so far, and can the estimate be paired with a range that reliably contains the true bill? Here the bill is a computed proxy, defined as recorded consumption in kWh multiplied by the published tariff rate, not a real invoice.

Accuracy is measured as the mean absolute error, in pounds, between each estimate and the bill calculated from the household's full recorded month. Reliability of the range is measured by how often that calculated bill falls inside it. To show whether a model adds value, its accuracy is compared against three simple guesses a household could make without one: continuing the daily rate recorded so far, assuming the bill matches last month's, or assuming it matches the same month one year earlier.

2 Project Objectives & Analytics Goal

The primary analytics task is supervised regression. For each household and month, the model will produce a point estimate of the month end bill proxy in pounds on day d, together with a prediction interval. This makes the task a form of probabilistic forecasting. On day d the consumption recorded so far is already known, so the model will predict only the consumption from day $d + 1$ to the end of the month and add it to the known part. The uncertainty therefore comes only from the remaining days, and the interval should narrow as the month progresses.

Such a model is only worth building if it improves on what a household could work out alone, so its estimates will be compared against three simple guesses that serve as benchmarks.

- **Run rate.** The rest of the month continues at the daily rate recorded so far.
- **Previous month.** The bill matches last month's bill.
- **Same month last year.** The bill matches the bill from one year earlier.

The project will build and test this model through four objectives.

1. Aggregate the half hourly readings into daily totals for each household and compute the monthly bill proxy for every household month.

2. Estimate the month end bill proxy on day d with a prediction interval, and compare it with the three benchmarks on later months that the model never saw during training.
3. Compare a single pooled model trained on all households with separate models trained on each household.
4. Define an alert rule that flags a household when its estimated bill exceeds a baseline by more than X percent.

If these objectives succeed, the model will beat all three benchmarks on mean absolute error, and its intervals will contain the calculated bill about as often as they claim to. Combined with the alert rule, it would warn a household early that its month is heading for an unusually high bill, while there is still time to change its usage.

3 Data Source Identification

The main source is the Kaggle release of the Low Carbon London smart meter data [2], which restructures the records published on the London Datastore [3]. Its half hourly consumption readings provide both the target and the main model inputs.

A household information file gives each household's tariff group and ACORN category, a UK consumer classification produced by CACI [4]. Most households paid a flat standard tariff, and approximately 1,100 paid a time of use tariff during 2013 [3]. The tariff group decides which rates apply, and the ACORN category may help a pooled model group similar homes.

Standard households paid 14.228 pence per kWh. During 2013, time of use households paid 3.99, 11.76 or 67.20 pence per kWh in Low, Normal and High periods, and a tariff schedule records which band applied in each half hour [3]. These rates turn recorded consumption into the bill proxy.

The Kaggle release also included hourly weather for the London area from the Dark Sky API and a list of UK bank holidays [2]. Temperature drives heating use, and holidays change daily routines, so both will serve as model inputs.

Source	Type	What it provides	Status
Half hourly consumption (Kaggle release of London Datastore data)	External, open, sensor	About 167M readings in kWh for 5,567 households, Nov 2011 to Feb 2014, CSV	Confirmed, downloaded
Household information and ACORN details	External, open	Tariff group and ACORN category for each household, CSV	Confirmed, downloaded
Published tariff rates	External, open	Flat rate and three time of use rates in pence per kWh	Confirmed, Datastore page
Tariff schedule (Tariffs.xlsx)	External, open	Band for each of the 17,520 half hours of 2013, Excel	Confirmed, downloaded and checked
Hourly weather (Dark Sky)	External, API derived	Hourly weather for the London area, included in the Kaggle release, CSV	Confirmed, downloaded
UK bank holidays	External, open	Dates of UK public holidays, CSV	Confirmed, downloaded

Table 1. Data source checklist.

4 Data Access, Quality & Feasibility

The team has already obtained every source in Table 1, so access is not a constraint. Rather than sample a subset, the project will aggregate all half hourly readings once into a table of daily totals for each household, which is small enough to model in memory.

Initial profiling of the raw data found no duplicate readings and no negative values. It also found three issues, which Phase 2 will address.

- **Clustered missing readings.** About 0.27% of expected half hourly readings are missing, and they mostly occur as whole days or longer with no data.
- **Zero runs.** 231 households record zero consumption for a day or longer, which could reflect meter faults or empty homes.
- **Late starts.** Most households joined between April and July 2012, so the same month last year benchmark is only available from mid 2013, which shortens the period available for testing.

The meter data is open under a Creative Commons Attribution 4.0 license [3]. The weather and ACORN data come from Dark Sky and CACI rather than UK Power Networks, so they carry their own terms, and the project uses only what the public release already contains [2]. Dark Sky also closed its API to new users after Apple acquired it in 2020 [5], so the weather data cannot be refreshed.

These sources cover everything the main task needs, from consumption to tariff rates, with no additional data required. The results may not transfer beyond London or to current prices, since the households form a trial sample and the rates date from 2013.

5 Preliminary Project Scope

The project is built for households that want an early estimate of their monthly bill. Energy suppliers and developers of household energy apps are secondary users, since they could build the approach into their own services.

To serve these households, the project will estimate month end bills for standard tariff homes at the flat rate, compare the model with the three benchmarks, compare pooled and per household models, and define the alert rule. Later phases will fix the day d, the alert threshold X, the alert baseline and the final model after testing several options for each. If time allows, a stretch goal will extend the analysis to time of use households in 2013.

Keeping that focus places several related goals out of scope. The project will not produce real invoices, so standing charges, VAT and actual billing cycles are excluded. Nor will it forecast demand for the grid or for London as a whole, deploy a real time system, or model the causal effect of weather on consumption.

Within this scope, the project will deliver four outputs.

- A reproducible pipeline that turns the raw readings into model ready data.
- A model that estimates the month end bill with a prediction interval, evaluated against the three benchmarks.
- Charts that show, for example households, how the estimated bill and its range narrow toward the actual bill as the month progresses.
- A final report and presentation that interpret the results for households, including how early in the month the estimate becomes reliable enough to act on.

References

[1] Yi Wang et al. “Review of Smart Meter Data Analytics: Applications, Methodologies, and Challenges”. In: *IEEE Transactions on Smart Grid* 10.3 (2019), pp. 3125–3148. DOI: 10.1109/TSG.2018.2818167.

[2] Jean-Michel D. *Smart Meters in London*. Kaggle. 2022. URL: https://www.kaggle.com/datasets/jeanmidev/smart-meters-in-london.

[3] UK Power Networks. *SmartMeter Energy Consumption Data in London Households*. London Datastore. 2014. URL: https://data.london.gov.uk/dataset/smartmeter-energy-use-data-in-london-households.

[4] Sagar Limbu. *What is Acorn?* CACI Limited. 2023. URL: https://acorn.caci.co.uk/what-is-acorn/.

[5] Greg Kumparak. *Apple Acquires Dark Sky, Android Version Shutting Down in July*. TechCrunch. Mar. 31, 2020. URL: https://techcrunch.com/2020/03/31/apple-acquires-dark-sky-android-version-shutting-down-in-july/.